

Lecture Notes: L-4

- **Wilcoxon Rank Sum Statistic (T):**

$$T = \sum_{i=1}^{n_1} R_{X_i} - \frac{n_1(n_1 + 1)}{2}$$

R_{X_i} is rank of X_i in the pooled sample. $X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}$

- **Mann-Whitney Statistic (T^*):**

$$T^* = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} I(X_i > Y_j)$$

- **Relationship:** Claim

$$T = T^*$$

- **Definitions:**

- R_{X_i} is the rank of X_i in the **pooled sample**:

$$\{x_1, \dots, x_{n_1}, y_1, \dots, y_{n_2}\}$$

- Ordered sample: $x_{(1)} < \dots < x_{(n)}$

- **Derivation:**

$$\begin{aligned} T &= \sum_{i=1}^{n_1} (R_{X_i} - i) \\ &= \sum_{i=1}^{n_1} (\#\{y_j < x_{(i)}\}) \quad (\text{Conceptually}) \end{aligned}$$

- **Visual Representation:**

$$x_1 \dots x_{n_1}, y_1 \dots y_{n_2}$$

- **Rank Definition Logic:**

$$R_{X_{(i)}} = \#\{samples < x_{(i)}\}$$

$$R_{X_{(i)}} = i + \#\{y_j < x_{(i)}\}$$

- **Equivalence:**

$$T = T^*$$

Where T^* is the **Mann-Whitney U Statistic**.

- **Properties of U-Statistic:**

- The U-statistic is a sum of unbiased estimators(in this case).
- Let $\theta = E[I(X_{(i)} > Y_{(j)})] = P(X > Y)$ of some parameter Q.
- $E[T^*] = n_1 n_2 \theta$.

- **Under Null Hypothesis (H_0):**

- $\theta = \frac{1}{2}$
- Expectation:

$$E[\hat{\theta}] = \frac{n_1 n_2}{2}$$

- Variance:

$$Var(\hat{\theta}) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

- **Asymptotic Normality:** A general procedure exists for finding the variance of U-statistics and establishing asymptotic normality.

$$Z = \frac{T^* - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \xrightarrow{d} N(0, 1)$$

(This is the two-sample test for center)

- **General form of Rank Statistics:**

$$\alpha \sum R_i + \beta$$

- **General Linear Rank Statistic (T_1):**

$$T_1 = \sum_{i=1}^{n_1} \phi \left(\frac{R_i}{N+1} \right)$$

where $N = n_1 + n_2$.

- **Assumptions on Score Function ϕ :**

$$- \int_0^1 \phi(u) du = 0 \implies E[T_1] = 0$$

$$- \int_0^1 \phi^2(u) du = 1 \implies Var(T_1) \approx 1 \text{ (Scale problem considerations)}$$

- **Measure of Scale:**

There is no unique measure of scale for a general distribution as there is median (alternative of mean in non-parametric)

- General measure of scale for a distribution:

$$\left(E[|X - \mu|^k] \right)^{1/k}$$

- $M = \text{Median}$
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- **Test for Comparing Scales of Two Populations:**

$$- \text{Let } X_1, \dots, X_{n_1} \sim F_X$$

$$- \text{Let } Y_1, \dots, Y_{n_2} \sim F_Y$$

- **Model (Location-Scale):**

$$\text{pdf of } X = f_X \left(\frac{x - \mu_1}{\sigma_1} \right)$$

$$\text{pdf of } Y = f_Y \left(\frac{x - \mu_2}{\sigma_2} \right)$$

Same form (f), different center and scale.

- **Hypothesis:**

- $H_0 : \sigma_1 = \sigma_2$
- $H_a : \sigma_1 > \sigma_2$

- **Fligner-Killeen (FK) Test Strategy:**

1. Estimate μ_1, μ_2 using sample medians.
2. Calculate absolute deviations:

$$u_i = |x_i - \hat{\mu}_1|$$

$$v_j = |y_j - \hat{\mu}_2|$$

if $\sum u_i \geq \sum v_i$ Then u_i is shifted to distribution of v_i

3. Perform **Wilcoxon Rank Sum Test** on $\{u_i\}$ and $\{v_j\}$.

Equivalent to taking squares, absolute values, or $\log |\cdot|$.

- **Notes:**

- Any monotonic increasing function can be used for the test.
- General rank statistics can be applied here with a different ϕ than the linear one.
- Example calculation mentioned: R_i is rank of $u_i = |x_i - \hat{\mu}_1|$ in the pooled sample. $S_\phi = \sum_{i=1}^{n_1} \phi(R_i)$
 - * Statistic $Z = 2.095$
 - * p -value = 0.036 (One-tailed)
 - * Conclusion: Reject H_0 (Need to use one-sided test).

- **Confidence Interval (CI):**

$$CI = (1.002, 5.63)$$

This is the CI for the ratio of scales $\frac{\sigma_2}{\sigma_1}$ (or σ_Y/σ_X).